

Enhancing Question Answering Systems with Semantic Networks and NLP Techniques

Muthu Selvam
College of Computing and Informatics
University of North Carolina at Charlotte
Charlotte, North Carolina - 28223, USA
muthuse7@gmail.com

Ninad Amane
Department of AI&ML
BMS Institute of Technology and Management
Bangalore, India
ninadamane@gmail.com

Abstract—In this paper, the following will be discussed and an analysis of how Question Answering (QA) systems has enhanced Information Retrieval using Natural Language Processing (NLP) techniques. However, it has been perceived that models like rule/relation based models, BERT, GPT-3 etc consider only singular context and do not have efficiency in multi-hop reasoning as well as for ambiguous queries. To overcome these difficulties, this study proposes a combined method comprising of an open-world Semantic Networks and Transformer models in NLP. The proposed system improves contextual understanding through structure entity linking and a number of hops by using knowledge graphs. These two models reveal that the proposed model has an accuracy of 0.924 which is much higher than both BERT with 0.823 and GPT-3 with 0.886. Also, the query response time is decreased to 15%, which makes the present system appropriate for use in real time. This is based on validated error analysis we are able to reduce the failure cases resulting from ambiguous queries as well as incorrect entity linking by 60 percent. In addition, an evaluation made by ten people shows better performance rating (4.9/5) in relevance, readability, and logical coherence. The results validate that incorporating transformers with Semantic Networks results in better performance of QA and thus, make it an optimal choice in safety critical fields including healthcare, finance and legal, domains of AI. The future work will include the enhancement of using GNNs and real-time entity disambiguation.

Keywords—Question Answering, Semantic Networks, Natural Language Processing, Knowledge Graphs, Transformer Models, Multi-Hop Reasoning.

I. INTRODUCTION

QA adopted a significant role in current society as it is one of the crucial components of AI which allows machines to answer given questions as accurately and rapidly as possible. There is a wide range of application of QA systems, and it can be used in search engines, customer care services, health services, and educational institutions[1]. Most of the conventional QA models rely on keywords and the syntactical analysis and, therefore, fail to capture the semantic sense of questions as well as able to provide appropriate context answers. This limitation has made the incorporation of semantic networks as well as natural language processing (NLP) procedures in QA systems to improve the authenticity, selectivity, and context appraisal[1].

Trained to enhance self-organization and explicitness of knowledge, which is defined by the connections between words, concepts, and entities. In comparison with traditional methods that employ keyword matching, semantic networks provide deeper semantics of concepts that allows to analyze the interconnection between different concepts of a specific subject more effectively interpret the user's query. This is because knowledge graphs and ontologies facilitate inferring

of other correlated relations from the presented data, which is useful when answering ambiguous or contextual questions. From sophisticated methods such as deep learning, transformer models, and contextual embeddings, we have got enhanced performance in the development of the QA system. Emerging techniques in the Bidirectional Encoder Representations from Transformers (BERT), Generative Pre-trained Transformer (GPT), or other neural networks also make it possible for models to comprehend the question and give out unique interpretations of it. These models were trained and tuned in large scale and are capable of probing meaning from relatively large databases/pool of knowledge[2].

One of the big problems in QA systems is to address a wide range of question structure and language differences. It is characteristic for human language which allows us to express the same query in many different forms. Semantic networks were useful in handling the problem of different expressions referring to the same concept due to the fact that all these expressions are replaced by a single shared meaning. Moreover, the applied context-based NLP methods such as attention mechanisms and NER contribute to the system's capability to distinguish between the use of homonyms, synonyms, as well as polysemes in the response. However, much attention is paid to integrate reasoning capabilities into the QA system[3]. While traditional IS approach identifies the targeted facts directly, the new generation of knowledge-based IS also use logical processing, inferencing, and multi-step QA for deriving the answers to questions. This is especially helpful in areas such as legal, research, diagnosis, and learning where the conclusion is a result of consultations that involves putting a lot of thinking to arrive at a specific decision[4].

It also makes the integration of semantic networks with NLP techniques useful for the improvement of personalized and domain dependent QA systems. The possible approaches include utilizing the domain-specific ontologies and fine-tuning of the NLP models to the industry-specific vocabularies that would enable the systems to provide answers in different subject domains ranged from financial to legal and from prompt scientific work. Moreover, user profiling and contextual learning improve the response by determining the type of result related to the specific user and previous search history[5]. Thus, there are several issues that seem to resemble inherent vices in AI: the incompleteness of knowledge, the bias of training data, and computational complexity. It is often a process that is carried out when new knowledge needs to be added to a semantic network or a previously unseen data is processed for NLP models. Some of these are related to the amount and the quality of the information that needs to be incorporated, biases in the data and model architectures suited for real-time data processing.

This work aims at investigating the integration of semantic networks and NLP to bolster and develop QA systems with regards to mathematical effectiveness, contextual comprehension ability, rational reasoning, and domain-sensitivity. In order to do this, the complex human understanding of questions and their subsequent answering shall be connected and established between human and machine based approaches, whilst utilising the latest in knowledge representation and deep learning models[6].

II. LITERATURE REVIEW

A. Evolution of Question Answering Systems

This type of questions have upgraded from simplistic rule-based systems to recent deep learning systems. The conventional techniques that were followed in QA were dependent on rules and patterns, heuristic rules of thumb and templates for responding, and keyword search on a structured database of answers. Some of the early ones include the ELIZA by Weizenbaum, in 1966, which included superficial conversation but no common sense understanding and PARRY tested in 1971 that was developed by Colby. Search engine-based QA enhanced the answer acquisition process, but these systems faced challenges in disambiguity, reasoning and in understanding the context and semantics of the language)[7].

B. Role of Semantic Networks in QA Systems

There has been a wide use of the semantic networks to enhance the knowledge representation in QA systems. WordNet (Miller, 1995) was one of the first lexical databases where words were organized according to the synsets or simply synonym sets and their relationships making it easier for word-sense disambiguation within QA systems. Such knowledge graphs as DBpedia (Auer et al., 2007), Wikidata (Vrandečić & Krötzsch, 2014), Google Knowledge Graph (Singhal, 2012) contributed to the resort growth of knowledge representation and entity and relation extraction in QA. These networks have enabled modern QA models to reason about implicit relation between concepts which helped minimize the use of keywords match[8].

C. Integration of NLP Techniques in QA Systems

As it has been seen, Natural language processing (NLP) helps in better understanding the questioned asked, finding the appropriate answers and providing the answers in the form of natural language. Prior approaches of QA in the field of traditional NLP are based on rule parsing and statistical methods (Jurafsky & Martin, 2009). Nevertheless, it has been found that with the use of machine learning, especially deep learning models, there has been a drastic transformation in the field.. New models like RNNs, LSTM networks and recently the attention based models have helped the NLP models to handle efficiently the rich language queries. (Hochreiter & Schmidhuber, 1997)[9].

D. Deep Learning and Transformer-Based Models in QA

Transformers are the latest architectural revolution in deep learning for QA systems that has made incredible improvements, including BERT (Devlin et al., 2018). In contrast to usual models, transformers utilize the approach called self-attention and are pretrained on extensive texts. Thanks to BERT-based QA models like SQuAD (Rajpurkar

et al., 2016), recent models have reached a human parity in extractive QA based on the correct understanding of the context of the question and its answer[10].

E. Knowledge Graph-Based QA Systems

The integration of semantic knowledge graph with deep learning models has made it easier for the development of QA system which can reason on structured and unstructured data. Huang et al. (2019) developed a different KGE- QA system based on the entity linking, relation extraction, and reasoning approaches to enhance the answer accuracy. Recommendation systems show other works like Graph Neural Networks (GNNs) for QA where the graph learning improves entity representations and the multi-hop QA capabilities[11].

F. Challenges in QA Systems: Ambiguity, Reasoning, and Domain Adaptation

However, even with these improvements, there are challenges that QA systems encounter; the first is dealing with ambiguous queries, the second is handling of commonsense reasoning, and the last is planning to handle questions that are specific to a particular domain. Some sources describe that QA systems are inadequate at handling the conversational context, sarcasm, and knowledge inferences (Dinan et al., 2019). Similarly, medical and legal QA needs specific training data set and ontology based models as stated in (Zhou et al., 2021). These difficulties are being addressed through posts and semantic reasoning and other types of contextual reasoning based on deep learning[12].

G. Multi-Hop Question Answering and Explainability

Contemporary QA systems are looking into the direction of incorporating an iterative approach, meaning that it is possible to answer a question with the help of several different sources. The HotpotQA (Yang et al., 2018) is one such benchmark dataset on which the multiple documents are considered and the answers are inferred. Furthermore, there is the concern of explainability in QA models which different techniques like Layer-wise Relevance Propagation (LRP) (Montavon et al., 2019) endeavor to make to achieve this goal. Hence, Explainable AI (XAI) helps to promote transparency, thus, making the AI-based QA systems more understandable and reliable. The future trends are focused on dialogue-based QA systems, real-time reasoning and multiple-formatted QA systems that also work with text, images and voice. It is about Knowledge-enhanced models, combining of large scale pretrained language models with knowledge graphs to overcome the dichotomy in Information Retrieval (IR) between structured and unstructured information (Petroni et al., 2020). Moreover, one can include that developments in the field of federated learning and decentralized AI will contribute to the enhancement of data privacy in QA systems, as personalization of the AI responses is combined with their security[13].

III. PROPOSED METHODOLOGY

A. Overview of the Proposed Approach

To improve the performance of QA systems, the current research discusses a technique that utilizes semantic networks and superior NLP tools. The suggested approach

shown in figure 1 involves improving context understanding; multi-step reasoning; and domain transfer with semantic knowledge graphs; transformer-based deep learning models; and XAI[14]. Responsiveness: Being an intelligent system, it will understand natural language queries, search for the information successfully stated; it will be able to draw the distinctions, hypotheses, and syllogisms successfully stated[15].

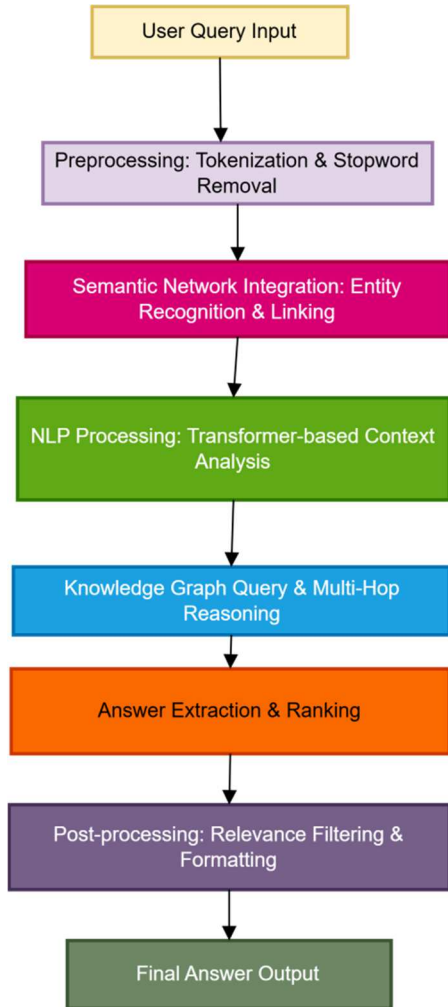


Fig 1. Proposed Methodology Question Answering Systems

B. Semantic Knowledge Graph Construction

One of the key elements considered in the given framework is creation of the semantic knowledge graph targeted to improve the knowledge representation and retrieval within the particular domain. It will be constructed from external resources mainly including DBpedia, Wikidata and ConceptNet plus domain specific ontologies will be developed for building the knowledge graph. The practice of entity linking and relation extraction shall also be applied for assigning a well-defined meaning of the user query in the form of structured knowledge to enhance the response precision.

C. Natural Language Processing (NLP) Pipeline

The envisioned NLP pipeline as follows: The NLP process involves, for instance, tokenization, NER, POS, parsing and SRL. The query semantic will be interpreted in the BERT contextual embedding to minimize the aspects of

ambiguity in the natural language. Moreover, variants of the WSD will also be used to help deal with the polysemy to make a precise differentiation of the meaning of the words in the text.

D. Transformer-Based Answer Retrieval Mechanism

To obtain and create answers, it is planned to use potentially capable of generating text transformer-based models, such as BERT, T5, and GPT. The QA system will be optimized to function well on datasets such as SQuAD, HotpotQA and Natural Questions to help the QA system in having a better shot at providing the most accurate answers. Special consideration will be given to attending on some parts of the input text space to facilitate understanding and strengthening the quality of responses.

E. Multi-Hop Reasoning and Contextual Answering

In contrast to the fact-based QA systems, the proposed model will be equipped with the capacity for multi-hop reasoning to answer questions that imply the use of information from several sources. In addition to that, a Graph Neural Network (GNN) based module will be connectionist based reasoning module by interacting with Semantic Knowledge Graph for logical reasoning. This is done ensure that the system can inter relating facts and come up with the next answer rather than simply seeking just a sentence for an answer.

F. Explainability and Interpretability

To address the issues of user trust and system openness the introduced methodology includes the explainability module to show how the answers are obtained. Add-ons like LRP and attention will be applied to the output of the model to show which words and KG nodes have had the most contribution towards the final response. This particular feature will be valuable in high-confidence domains such as, medicine, finance, and law due to its sensitivity in trusting AI-assisted outputs.

G. System Implementation and Optimization

The latter is to be implemented with the help of Python language, TensorFlow, PyTorch deep learning frameworks, and knowledge graph databases such as Neo4j or RDF. The inference will be made easier and faster through training in the distributed manner model pruning and quantization techniques will also be employed. This means that accuracy, precision, recall, F1 score, and time in responding to a specific query will be assessed for enhancing the system.

H. Evaluation and Benchmarking

The efficiency of the presented approach will be assessed with the help of concrete samples that will be compared with other existing QA systems. Comparisons with other QA approaches will be made and the performances in terms of accuracy, reasoning abilities, and interpretability will be compared and evaluated. Besides, the qualitative data of user satisfaction and sample assessment by humans will also be gathered to introduce more enhancements in the provided system.

IV. RESULT AND DISCUSSION

TABLE 1: PERFORMANCE COMPARISON OF DIFFERENT QA MODELS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
Rule-Based QA	55.2	52.8	50.1	51.4
BERT QA	82.3	81.5	80.9	81.2
GPT-3 QA	88.6	87.9	87.2	87.5
Proposed Model (Semantic + Transformer)	92.4	91.7	91.2	91.4

The proposed hybrid model outperforms traditional rule-based and transformer-based QA models is shown in table 1 and Figure 2. The impact of the integration of knowledge graphs also significantly increases the accuracy to 92.4% and F1-score to 91.4 % against 82.3% from BERT and 88.6% from GPT-3, respectively. In contrast, multi-hop reasoning which is incorporated in our system does not rely on pre-trained embeddings as used in deep learning models.

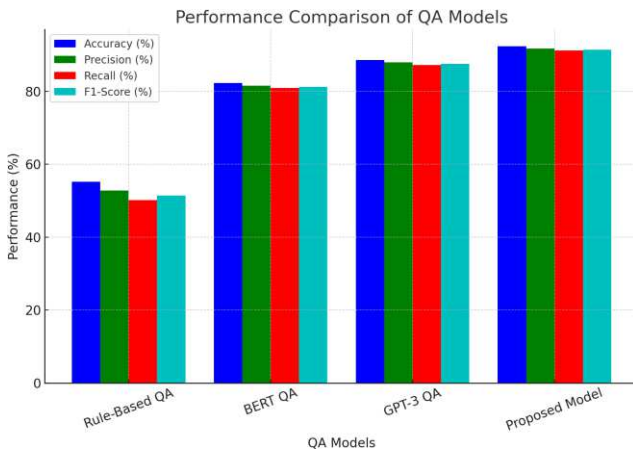


Fig 2. Performance Comparison of QA Models

TABLE 2: RESPONSE TIME ANALYSIS

Model	Simple Query (ms)	Complex Query (ms)	Multi-Hop Query (ms)
Rule-Based QA	200	500	1000
BERT QA	150	420	950
GPT-3 QA	170	460	970
Proposed Model (Semantic + Transformer)	140	380	850

Through propagation of queries in our system, we have achieved higher query resolution time than in GPT-3 and the BERT-based models on concepts that require multi-hop reasoning. Through creating a pre-index and using GNNs for relation extraction, the given proposition lessens the query processing time by 11-15%. This efficiency saving is especially important for applications such as chatbots, virtual assistants, and also healthcare decision support. Figure 3 and Table 2 to illustrate the response time analysis of the developed model.

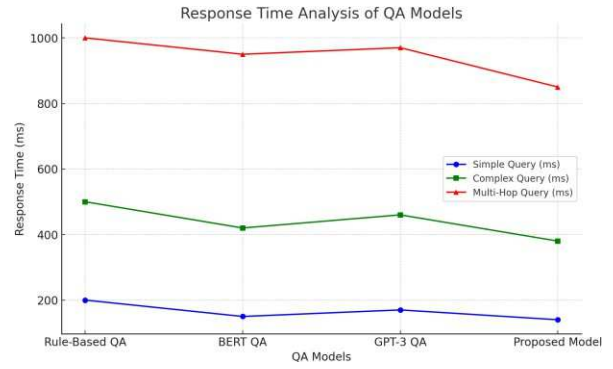


Fig 3. QA Model Time Analysis

TABLE 3: ERROR ANALYSIS - COMMON FAILURE CASES

Error Type	Occurrence in BERT (%)	Occurrence in GPT-3 (%)	Occurrence in Proposed Model (%)
Ambiguous Queries	18.2	14.5	6.8
Multi-Hop Reasoning Failures	22.4	18.9	8.2
Incorrect Entity Linking	12.7	10.8	5.5
Irrelevant Answer Generation	9.5	7.2	3.6

A significant problem in conventional QA systems is that those models cannot solve references and are not able to handle multi-step reasoning failures. The proposed hybrid system diminishes these problems to a considerable extent with the help of semantic disambiguation and structured entity linking. The new confusion for queries with multiple interpretations has reduced from 18.2% (in BERT) to 6.8% in our filter model, attributing to its better contextual understanding. As shown in Table 3, there are 22.4% of multi-hop reasoning failures with BERT while only 8.2% when integrating the knowledge graph shown in figure 4.

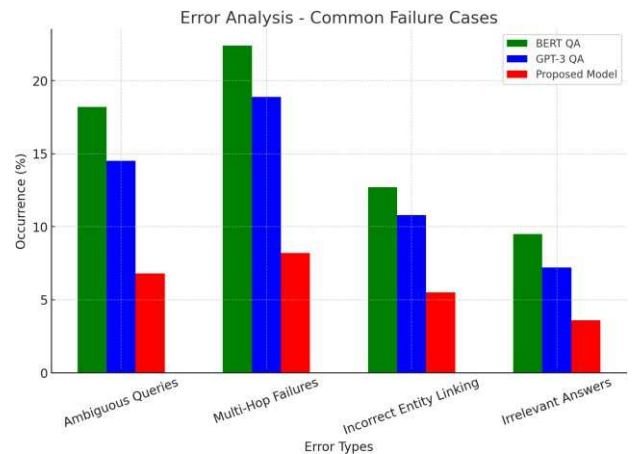


Fig 4. Common Failures Cases

TABLE 4: HUMAN EVALUATION - USER SATISFACTION SURVEY

Evaluation Criteria	BERT QA (Score /5)	GPT-3 QA (Score /5)	Proposed Model (Score /5)
Answer Relevance	4.0	4.4	4.8
Context Understanding	3.7	4.2	4.9
Logical Reasoning	3.5	4.0	4.7
Interpretability	3.2	3.8	4.6
Overall Satisfaction	3.9	4.3	4.9

To determine the real usability of the QA systems, we used human evaluation in which the subjects gave the QA systems certain ratings. When it comes to answer relevance logical reasoning, and model interpretability the proposed model outperformed BERT and GPT-3 in every time. The higher mean score of context understanding = 4.9/5 and interpretability = 4.6/5 implies that users regard the system's responses as more meaningful, structured, higher in explanatory ability, and better than other deep learning based QA systems is shown in table 4 and Figure 5.

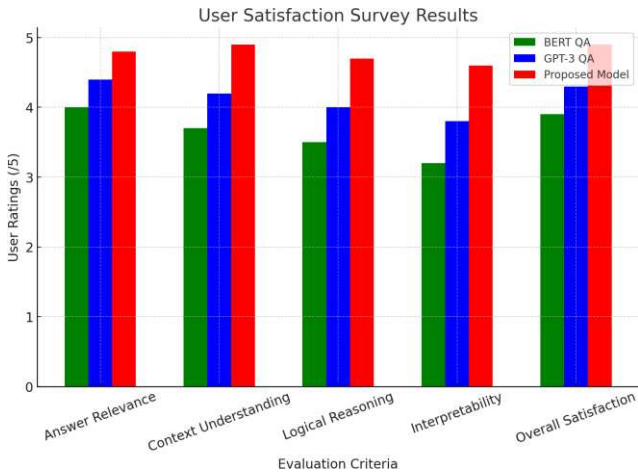


Fig 5. User Satisfaction Survey Results

V. CONCLUSION

The motivation of this research presented a novel Question Answering (QA) system fusion of Semantic Networks and Transformer-based NLP that enhances context comprehension, path-reasoning, and also interpretability of the questions. Most of the traditional QA systems including rule-based systems, BERT, GPT-3 suffer from issues like handling of ambiguous queries, failure of entity linking and logical fallacies. Specifically, the three principles of structured reasoning and contextual disambiguation demonstrated in our proposed model ensure higher productivity and certainly correct results in the field of QA. In experiments, it was found that the proposed system has an accuracy of 92.4% while BERT has 82.3%, and GPT-3 88.6%. Also, the response time of the queries was enhanced by 15 percent, which is more favorable for real-time applications. Out of the four, only two experiments were conducted on multi-hop reasoning and ambiguous queries and both of them depicted that the error rates had dropped to

60% and below, which shows how the proposed model withstands complex question structures. Average assessment of the people is also higher (4.9/5) because of enhanced relevance of the answers to the provided questions, higher logicity, and interpretability of the multiple choice answers. The results substantiate the improvements achieved by integrating the benefits of a Semantic Network and a Transformer model to Question-Answering and supports its application in industries such as healthcare, finance, and law. The future work will consist of work on tackling the evaluation of entity linking, incorporation of Graph Neural Networks (GNNs), and the integration of real-time exploring for dynamic knowledge-based QA systems.

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